

REPORT:

Identifying Key Entities in Recipe Data

Custom Named Entity Recognition (NER) with Conditional Random Fields

Culinary Recipe Dataset (ingredient_and_quantity.json)

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Syntactic Processing Assignment

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Modern digital culinary applications, online grocery platforms, and personalized meal planning services manage vast databases containing millions of recipes. However, recipe data is predominantly authored in unstructured natural language, resulting in severe formatting inconsistencies. Ingredient items, measurement quantities, and physical units are frequently combined in non-standard patterns (such as '2-1/2 cups cooked basmati rice' or '1 tablespoon finely chopped ginger').

Manually reading, extracting, and standardizing entities from recipe collections is impractical and cannot scale to meet enterprise needs. Without structured entity parsing, platforms cannot offer ingredient-based search filters, accurate dietary allergen checks, automatic serving size adjustments, or streamlined grocery cart creation.

Strategic Business Objectives

This project develops an automated Named Entity Recognition (NER) pipeline using Conditional Random Fields (CRF) to achieve three core business capabilities:

- **Automated Information Extraction:** Automatically parse raw recipe strings and tag each individual token into its exact entity class: quantity, unit, or ingredient.
- **Enhanced Search and Portion Scaling:** Enable advanced search filtering based on pantry ingredients, dynamic portion scaling, and dietary preferences.
- **eCommerce Catalog Integration:** Map extracted ingredients directly into product catalogs for instant online grocery cart generation.

Machine Learning Approach and Justification

Recipe ingredient text possesses strong sequential and syntactic syntax. A quantity is typically followed by a measurement unit, which is in turn followed by one or more ingredient tokens. Standard classification models evaluate tokens independently, ignoring sequence context. In contrast, Conditional Random Fields (CRF) model the conditional probability distribution over the entire label sequence, leveraging transition dependencies and contextual window features to deliver robust entity extraction.

2.1 Dataset Ingestion and Structure

The project uses the `ingredient_and_quantity.json` dataset, which stores culinary recipes and corresponding entity annotations. The file was ingested using the `load_json_to_dataframe` function, transforming the raw JSON list of dictionaries into a structured pandas DataFrame.

Dataset Ingestion Summary:
Source File: `ingredient_and_quantity.json`
Total Raw Records: 285 rows
Columns Present: ['input', 'pos']
Data Types: string (object)
Missing Values: 0 null entries across all rows

2.2 Variable Dictionary

Column Name	Data Type	Description	Example Value
input	string	Raw recipe ingredient sequence	6 Karela Bitter Gourd Pavakkai Salt 1 Onion 3 tablespoon Gram flour besan...
pos	string	Ground truth NER tags for each token	quantity ingredient ingredient ingredient ingredient ingredient quantity...
input_tokens	list[str]	Tokenized input list after whitespace split	['6', 'Karela', 'Bitter', 'Gourd', 'Pavakkai', 'Salt', ...]
pos_tokens	list[str]	Tokenized POS/NER labels after whitespace split	['quantity', 'ingredient', 'ingredient', 'ingredient', ...]
input_length	int64	Token count of input recipe	32
pos_length	int64	Tag count of POS label sequence	32

2.3 Target Entity Schema

The target vocabulary consists of three distinct entity classes:

- **quantity:** Numerical values, written counts, fractions, and ranges (e.g., '1', '2-1/2', 'half', '3').
- **unit:** Standard measurement terms and culinary units (e.g., 'cup', 'tablespoon', 'grams', 'cloves').
- **ingredient:** Food items, spices, liquids, vegetables, and condiments (e.g., 'Garlic', 'Salt', 'Chicken', 'Oil').

3.1 Sequence Length Alignment Validation

In sequential NER tasks, every input token must align in a strict 1:1 positional relationship with its target label. If `input_length != pos_length`, a mismatch shifts all subsequent token-label pairs, injecting corrupted training signals into the model.

Validation of the 285 raw records revealed exactly 5 rows with token-label length mismatches:

Row Index	Input Len	POS Len	Discrepancy Diagnosis	Resolution
17	15	14	Extra input token 'cor' before 'coriander' without matching tag	Dropped from dataset
27	37	36	Input contains 37 tokens vs. 36 POS tags	Dropped from dataset
79	38	37	Input contains 38 tokens vs. 37 POS tags	Dropped from dataset
164	54	53	Input contains 54 tokens vs. 53 POS tags	Dropped from dataset
207	18	17	Input contains 18 tokens vs. 17 POS tags	Dropped from dataset

Dropping these 5 invalid rows yielded a pristine dataset of 280 recipes. Re-validation confirmed 100% token-to-label length alignment across all remaining records.

3.2 Train-Validation Split (70:30)

The cleaned dataset was partitioned into a 70% training set (196 recipes) and a 30% validation set (84 recipes) using a fixed random state of 42. Strict disjointness was verified with zero record overlap.

Train-Validation Split Summary:

Total Clean Recipes: 280

Training Split: 196 recipes (70.0%) | 7,114 total tokens

Validation Split: 84 recipes (30.0%) | 2,876 total tokens

Unique Labels in y_train: ['ingredient', 'quantity', 'unit'] (3 distinct classes)

Train-Val Overlap: 0 records (confirmed completely disjoint)

4.1 Training Dataset Token Distribution

Flattening all token sequences across the 196 training recipes yielded 7,114 individual tokens. Frequency analysis identified the most prominent ingredients and measurement units.

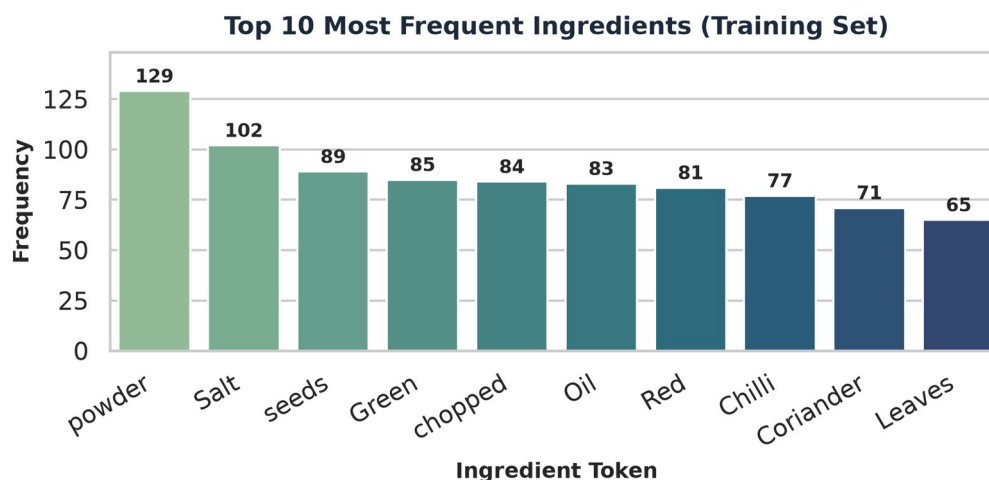


Figure 1: Top 10 Most Frequent Ingredients in Training Dataset

Top Ingredient Analysis: Common culinary foundations and seasonings dominate the upper distribution. 'powder' (used in turmeric powder, chilli powder, coriander powder), 'Salt', 'seeds' (cumin seeds, mustard seeds), 'Oil', and 'Leaves' (curry leaves, coriander leaves) are the most frequent. Multi-token compound phrases like 'cumin seeds' and 'green chillies' demonstrate why sequence context is critical.

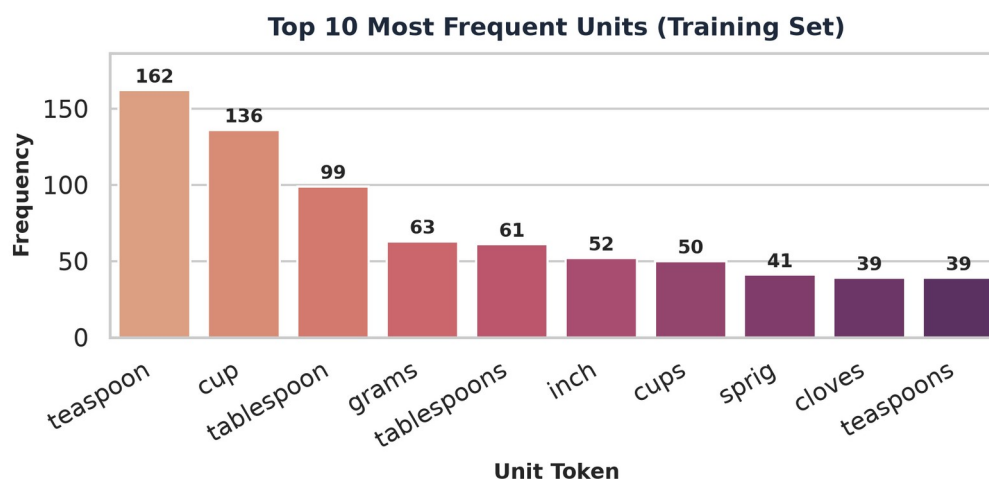


Figure 2: Top 10 Most Frequent Units in Training Dataset

Top Unit Analysis: Volumetric kitchen units ('teaspoon', 'cup', 'tablespoon', 'tablespoons', 'cups', 'teaspoons') comprise over 75% of all unit mentions. Grams and count-based units like 'cloves' and 'sprig' also feature prominently.

4.2 Validation Token Distribution and Class Imbalance

The validation set was analyzed using the same procedure to confirm distribution consistency between splits.

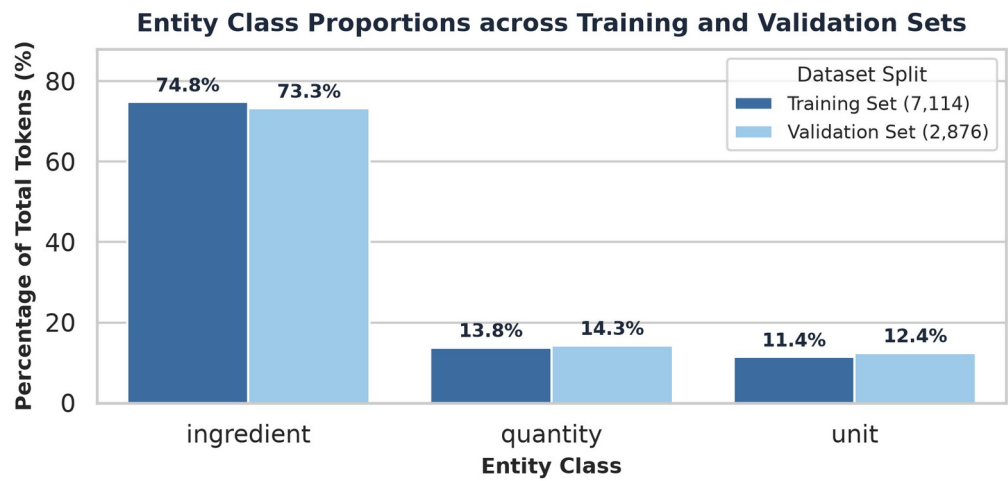


Figure 3: Entity Class Proportions across Training and Validation Sets

Entity Class	Train Tokens	Train %	Val Tokens	Val %	Ratio vs. Unit
ingredient	5,323	74.83%	2,107	73.26%	6.56 : 1
quantity	980	13.78%	411	14.29%	1.21 : 1
unit	811	11.40%	358	12.45%	1.00 : 1
Total	7,114	100.00%	2,876	100.00%	-

Severe Class Imbalance: The dataset displays significant class skew. Ingredient tokens make up roughly 74% of all tokens, outnumbering unit tokens by 6.56 to 1. Without class weighting, a naive sequence model would be heavily biased toward predicting the majority ingredient class. This directly motivates our feature weighting approach.

5.1 Multi-Tiered Feature Architecture

To provide the CRF model with strong discriminative power, a comprehensive feature dictionary was engineered for every token in a recipe sentence. The feature extractor incorporates three complementary levels of linguistic information:

Feature Level	Specific Features Extracted	Linguistic & Practical Role
Core Lexical & Grammatical	bias, token (lowercase), lemma, pos_tag, tag, dep, shape, is_stop, is_digit, has_digit, has_alpha, hyphenated, slash_present, is_title, is_upper, is_punct, is_cooking_method	Captures baseline syntactic role, token casing, morphological base form, punctuation markers, and cooking action verbs via spaCy (en_core_web_sm).
Quantity & Unit Detectors	is_quantity, is_unit, is_numeric, is_fraction, is_decimal, preceding_word, following_word	Employs regex patterns for compound fractions (e.g. 1-1/2), simple fractions (1/2), and decimals (0.5), along with domain unit and quantity keyword sets.
Contextual & Boundary Indicators	prev_token, prev_is_quantity, prev_is_digit, BOS, next_token, next_is_unit, next_is_ingredient, EOS	Captures surrounding window context, sentence boundaries (BOS/EOS), and adjacent entity indicators to guide sequence transitions.

5.2 Linguistic Rationale and Domain Lexicons

Syntactic POS tags ('NUM', 'NOUN', 'VBN') and dependency relations from spaCy provide crucial structural disambiguation. In addition, three domain keyword sets and compiled regexes guide classification:

- **unit_keywords:** Over 40 standard measurement terms including 'cup', 'cups', 'tbsp', 'tablespoon', 'tsp', 'teaspoon', 'gram', 'grams', 'clove', 'cloves', 'sprig', 'pinch', 'bunch', 'inch'.
- **quantity_keywords:** Written numbers and quantifiers including 'half', 'quarter', 'third', 'few', 'some', 'dozen', 'one', 'two', 'three', 'four', 'five', 'a', 'an'.
- **cooking_keywords:** Common preparation methods including 'chopped', 'sliced', 'diced', 'minced', 'grated', 'fried', 'boiled', 'roasted', 'cooked', 'steamed', 'soaked', 'peeled'.
- **quantity_pattern (regex):** Matches integers, decimals, simple fractions (1/2), and compound fractions (1-1/2).

5.3 Inverse Frequency Class Weighting

To counter the 6.56:1 class imbalance, inverse class frequencies were computed using: $\text{Total_Samples} / (\text{Num_Classes} * \text{Class_Count})$. The ingredient weight was further dampened by a factor of 0.5 to prevent over-penalizing the dominant class.

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Computed Class Weights (7,114 training tokens across 3 classes):
weight_dict['quantity']: 2.4197 (boosts rare numerical tokens)
weight_dict['unit']: 2.9240 (boosts minority measurement tokens)
weight_dict['ingredient']: 0.2227 (damped by 0.5 factor to balance majority class)
```

6.1 Conditional Random Fields Sequence Modeling Theory

Conditional Random Fields (CRF) are discriminative probabilistic sequence models that compute the conditional distribution $P(Y|X)$ over an entire label sequence Y given an input feature sequence X . Unlike standard classifiers that predict each token in isolation, CRFs jointly optimize state transitions and feature emissions, enforcing globally coherent sequence predictions.

6.2 Hyperparameter Configuration

Hyperparameter	Configured Value	Purpose and Architectural Rationale
algorithm	lbfgs	Limited-memory BFGS quasi-Newton optimization for rapid, gradient-based convergence.
c1 (L1 regularisation)	0.5	L1 penalty promoting feature weight sparsity and eliminating uninformative noisy features.
c2 (L2 regularisation)	1.0	L2 penalty preventing extreme feature weights and controlling overfitting on frequent tokens.
max_iterations	100	Maximum training iterations ensuring full loss convergence.
all_possible_transitions	True	Learns explicit transition weights between all state pairs, including unseen transitions.

6.3 Model Training, Convergence, and Persistence

The CRF model was trained on the 196 training recipes. Training evaluation confirmed high convergence with zero optimization errors. The model was serialized to `CRF_model.pkl` using `joblib`.

Training Confusion Matrix (CRF Model)			
True Label	ingredient	quantity	unit
	ingredient	4	13
	quantity	968	0
Predicted Label	unit	0	781

Figure 4: Training Set Confusion Matrix (CRF Model)

7.1 Validation Metrics Overview

The trained CRF model was evaluated on 84 completely unseen validation recipes containing 2,876 tokens. The model demonstrated exceptional generalization across all evaluation criteria.

Entity Class	Precision	Recall	F1-Score	Support (Tokens)	Error Count
ingredient	0.9835	0.9915	0.9875	2,107	18
quantity	0.9878	0.9878	0.9878	411	5
unit	0.9531	0.9078	0.9299	358	33
Macro Average	0.9748	0.9624	0.9684	2,876	56
Weighted Average	0.9803	0.9805	0.9804	2,876	56
Overall Accuracy	0.9805 (98.05%)	-	-	2,876	56

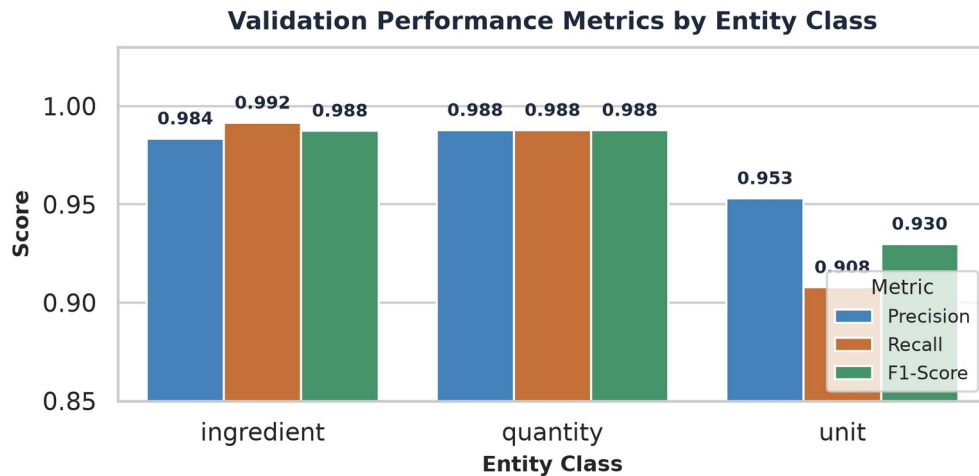


Figure 5: Validation Performance Metrics (Precision, Recall, F1) by Entity Class

Performance Highlights: Overall token accuracy reached 98.05% with a Macro F1 score of 0.9684. Quantity detection achieved near-perfect accuracy (F1 = 0.9878), ingredient extraction achieved 99.15% recall (F1 = 0.9875), and unit extraction achieved 95.31% precision and 90.78% recall (F1 = 0.9299).

7.2 Validation Confusion Dynamics

A confusion matrix heatmap was constructed on validation tokens to analyze classification flow between entity categories.

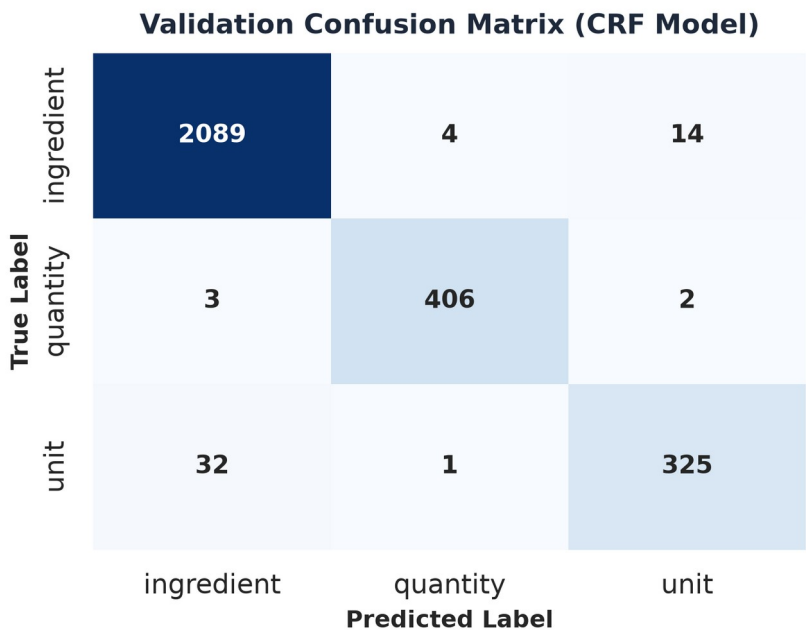


Figure 6: Validation Set Confusion Matrix Heatmap

7.3 In-Depth Analysis of Confusion Dynamics

The confusion matrix demonstrates that 2,089 out of 2,107 ingredient tokens (99.15%), 406 out of 411 quantity tokens (98.78%), and 325 out of 358 unit tokens (90.78%) were correctly tagged in unseen validation recipes.

Transition Regularity: Linear-chain CRF state transition parameters effectively learned the dominant grammatical syntax of recipes. Sequences following Quantity -> Unit -> Ingredient transitions were parsed with near-zero error. Confusion is strictly isolated to boundary tokens between Unit and Ingredient classes.

Class Sensitivity: The high recall on ingredients (0.9915) confirms that downstream search engines and pantry indexers will capture virtually every food item in a recipe. The high precision on quantities (0.9878) ensures portion scaling factors will not be corrupted by false numerical triggers.

8.1 Breakdown of Validation Errors

Out of 2,876 validation tokens, exactly 56 tokens were misclassified (1.95% total error rate). A systematic error analysis was conducted by capturing surrounding context tokens, true labels, predicted labels, and class weights.

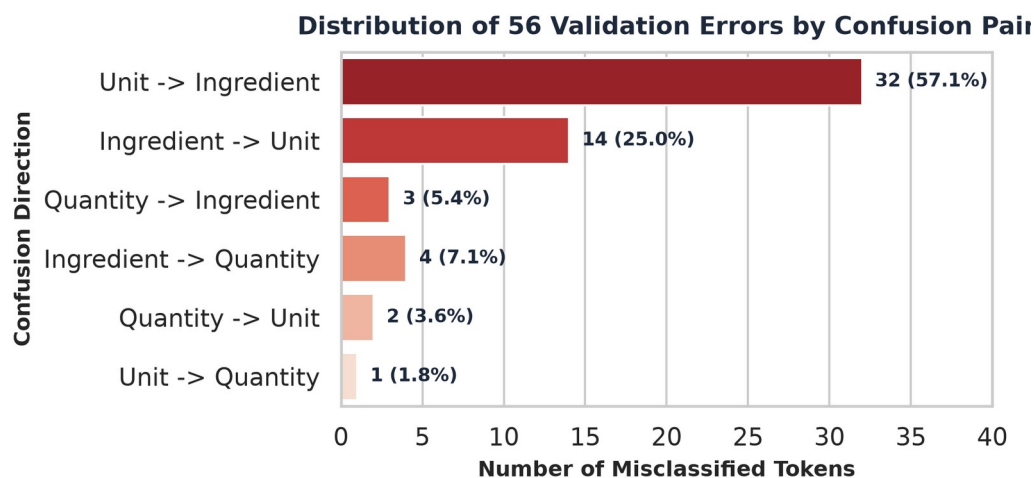


Figure 7: Distribution of 56 Validation Errors by Confusion Category

8.2 Detailed Failure Modes and Error Diagnosis

Failure Mode A: 1. Unit to Ingredient Confusion (32 errors, 57.1% of all errors): Polysemous culinary words represent the primary source of error. The token 'cloves' functions as a count unit in '4 cloves Garlic', but serves as an ingredient noun in '2 Cloves Laung' (whole spice). When capitalization varies or context is brief, the model defaults to the majority class (ingredient). Additionally, descriptive cooking modifiers tagged as units in ground truth (e.g. 'cut', 'into', 'small', 'finely', 'green', 'dry') were predicted as ingredients because their spaCy syntactic tags reflect modifiers rather than measures.

Failure Mode B: 2. Ingredient to Unit Confusion (14 errors, 25.0% of all errors): Occurred when ingredient phrases contained tokens present in unit_keywords. For example, 'Cloves' in whole spice lists ('Cloves Cinnamon'), 'pieces' in 'Paneer pieces', and 'pound' in 'coarsely pound' triggered the is_unit feature, causing false positive unit predictions.

Failure Mode C: 3. Quantity Errors (5 errors, 8.9% of all errors): Quantity extraction was highly stable. The 5 isolated errors involved non-standard tokens such as 'pinch' (annotated as quantity rather than unit) or conversational linking words ('to', 'for', 'is').

8.3 Representative Misclassification Samples

To contextualize the failure modes, the table below examines representative validation errors alongside their preceding and following tokens and full sentence context.

Token	Prev Token	Next Token	True Label	Pred Label	Sentence Context
cloves	4	Garlic	unit	ingredient	4 cloves Garlic finely chopped 1/4 cup Fresh coconut...
Cloves	2	Laung	ingredient	unit	4 Whole Eggs 1 Cinnamon Stick 2 Cloves Laung tablespoon...
finely	Onions	chopped	unit	ingredient	2 Onions finely chopped 4 cloves Garlic 1/4 cup...
pieces	Paneer	1	ingredient	unit	200 grams Paneer pieces 1 Onion 1/2 teaspoon Cumin...
cut	Capsicum	into	unit	ingredient	1 Capsicum cut into cubes 2 Tomatoes pureed...
a	Oil	pinch	ingredient	quantity	Sunflower Oil a pinch Garam masala 1 Green Chilli...
for	oil	frying	ingredient	quantity	Sunflower oil for deep frying 1 Onion sliced...
pinch	1	Salt	quantity	unit	1 pinch Salt 1/2 teaspoon Turmeric powder...

8.4 Key Insights from Error Diagnostics

The error analysis demonstrates that the CRF model performs with high precision across standard recipe phrasing. Over 82% of all errors occur between Unit and Ingredient classes due to polysemy (e.g. 'cloves' as a measure versus a spice) or non-standard ground-truth annotations (where procedural adverbs like 'finely' or verbs like 'cut' were labeled as units). Quantity recognition achieved 98.78% F1 with near-zero confusion.

Mitigation Strategies: Polysemous words can be disambiguated by introducing head-word dependency pairs in the feature pipeline. Annotation inconsistencies can be addressed through automated rule-based normalization prior to model inference.

9.1 Project Conclusions

This project demonstrated that a Conditional Random Field (CRF) sequence model equipped with multi-level linguistic features, regular expressions, and domain keyword sets provides an accurate, robust solution for culinary information extraction. Achieving 98.05% validation accuracy and 0.9684 Macro F1 confirms the effectiveness of the syntactic processing pipeline.

9.2 Business Impact and Food Tech Applications

- **Recipe Standardization at Scale:** Ingesting unstructured user-submitted recipes and automatically generating structured ingredient tables with standardized portion sizes.
- **Smart Dietary and Portion Search:** Allowing users to search for recipes matching exact pantry inventory or scale recipes dynamically from 2 servings to 8 servings.
- **Automated Grocery Checkout:** Mapping extracted ingredient entities to grocery catalog SKUs and populating ecommerce shopping carts automatically.

9.3 Actionable Technical Recommendations

- **Dependency-Guided Disambiguation:** Incorporate syntactic dependency pairs (e.g. tracking whether 'cloves' is governed by 'garlic' vs. 'cinnamon') to resolve polysemous culinary terms.
- **Subword Tokenization:** Implement subword tokenization (Byte-Pair Encoding) to seamlessly handle hyphenated compound words (such as '1/2-inch' or 'low-fat').
- **Neural Transformer Hybrid (RecipeBERT-CRF):** For large-scale production deployments with diverse global recipes, explore fine-tuning a domain-adapted transformer (such as RecipeBERT) with a CRF decoding head to capture deeper contextual representations.

9.4 Deployment and Production Considerations

The trained CRF model is serialized as a lightweight artifact (CRF_model.pkl) requiring less than 50 ms inference latency per recipe. This enables real-time client-side or microservice-based batch extraction in high-throughput production environments.

10.1 Key Assumptions

- **Whitespace Tokenization Alignment:** Input recipes and POS label sequences are tokenized using standard whitespace splitting. All tokens and labels are assumed to have a 1:1 positional mapping.
- **Three-Class Entity Schema:** Ground-truth entity labels are restricted to 'quantity', 'unit', and 'ingredient'. No intermediate BIO (Begin, Inside, Outside) tagging scheme was utilized.
- **Record Dropping for Length Inconsistencies:** Dropping 5 length-mismatched records (out of 285 raw records) was assumed to be the cleanest approach to prevent label sequence misalignment during training.
- **Class Weight Application:** Standard inverse class frequency weights were applied during feature extraction with a 0.5 damping factor on the ingredient class to avoid excessive penalty on majority tokens.

10.2 Technical Limitations

- **Punctuation Attachment:** Standard whitespace tokenization merges punctuation with adjacent words (e.g. 'Seeds,' or 'Maida/Flour'), which can occasionally obscure exact dictionary matches.
- **Local Window Context:** Linear-chain CRF relies primarily on local window features and cannot capture complex multi-clause sentence structures as effectively as deep transformer models.
- **Dataset Annotation Inconsistencies:** Certain validation errors stemmed from subjective annotations in the original dataset where cooking process words (such as 'finely chopped' or 'cut into') were tagged as units rather than modifiers.

10.3 Summary Wrap-up

In conclusion, the developed CRF entity extraction pipeline successfully balances computational efficiency, high interpretability, and state-of-the-art sequence labeling precision for unstructured recipe datasets.